START

N

Natural numbers

the counting numbers 1, 2, 3, ...

`subtraction demands answers beyond the counting numbers`

1ST EXTENSION

Z

Integers

the negatives join the counting numbers

`division insists on exactness`

2ND EXTENSION

Q

Rational numbers

fractions: every division now has an exact result

`geometry requires lengths no fraction can capture`

3RD EXTENSION

R

Real numbers

the irrationals fill the line

`$x^2 + 1 = 0$ has no real solution`

4TH EXTENSION

C

Complex numbers

a new unit i with $i^2 = -1$, perpendicular to the real line

EACH TIME declare the problem unsolvable, or expand what counts as a number. Mathematics chose to expand.

N → Z → Q → R → C

Select a level for details.